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disassembler: planetary-scale sequence extraction from Logan

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Logan indexes the SRA petabase as **unitig graphs** and **assembled contigs**. Given one reference CDS, `disassembler` reconstructs full-length haplotypes from either—ending in the same **codon MSA, haplotype catalog, and tree**.

Galaxy PROJECT

logan-disassembler
github.com/nekrut/disassembler — Graph path-walker & contig streamer over Logan.

A Two routes, one cohort shape

A graph path-walker on Logan unitigs and a streaming assembler on Logan contigs, scanning the same planetary SRA petabase from one reference CDS.

LOGAN-DISASSEMBLER · HyphAeon · PennState · INSTITUT PASTEUR · TEMPLE UNIVERSITY · NBIS

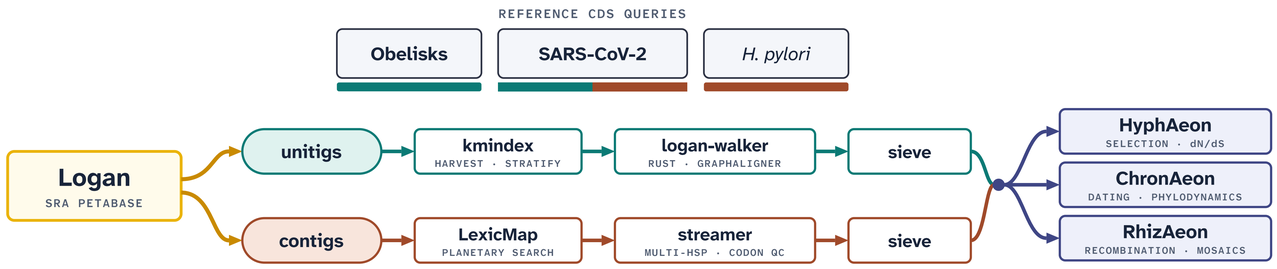
LOGAN UNIVERSE
4.99 M
subject genomes in the LexicMap index—the universe each query scans.

WALKER CONTAMINATION FILTER
0.555% → 0%
mislabeled spike accessions, raw unitig hits → `CLEAN_FULL_LENGTH`.

ROUTE A · UNITIG GRAPH

When the walker route wins

- **Shattered genes.** Zika NS5: ~50 contig fragments, one clean walk.
- **Divergent tails** a single-reference search never reaches.
- **Intra-host variation** kept, not collapsed: per-site allele frequencies.



ROUTE B · CONTIG STREAMER

When the LexicMap route wins

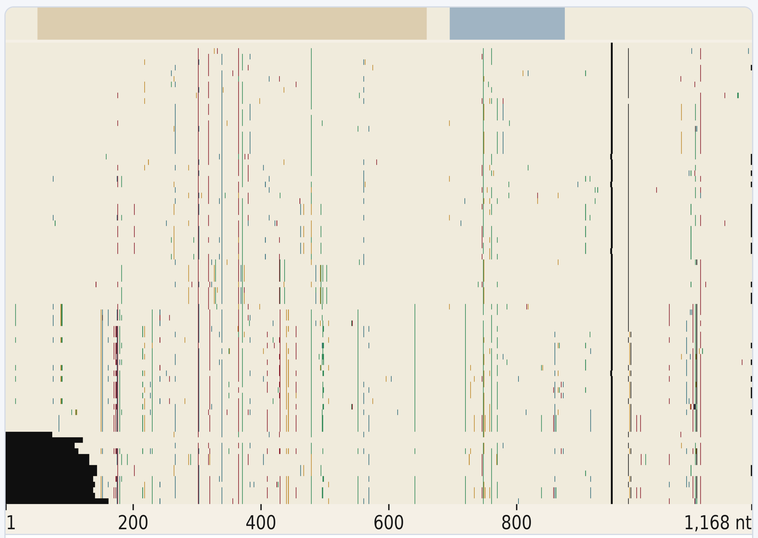
- **Split genes.** Collinear tiling stitches a 55/45 HSP pair that fails `qcovHSP ≥ 80`.
- **Planetary scale.** 24 GB streamed; memory is *O*(haplotypes), not *O*(rows).
- **Hands off to the walker** below 85% median identity.

B Obelisks across microbiomes

ROUTE A · WALKER

METATRANSCRIPTOMIC_BCT dominates: obelisks live in bacterial microbiome transcriptomes across human and microbial cohorts.

KMINDEX HITS	MAX CONTAIN.	SHARDS QUERIED	CENTROIDS
924	0.97	141	5
unique runs	alpha	meta + metaT	α β γ ε Ss



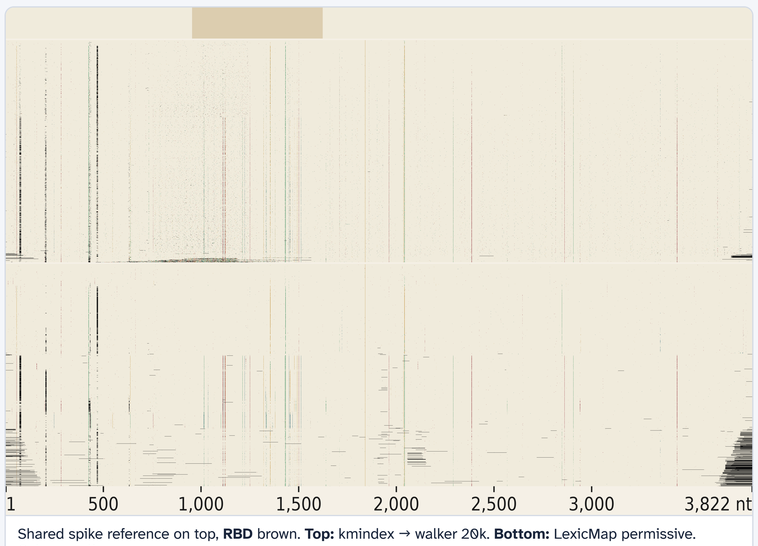
Obelisk-*alpha*. Reference strip on top: **Oblin-1** brown, **Oblin-2** blue. Rows sorted by edit count.
match A C G T gap

C SARS-CoV-2 spike: two routes

ROUTE A ROUTE B

The routes are complementary. Only the walker touches the divergent tail; only LexicMap-permissive catches 1.15 M Alpha/Delta-era runs strict loses.

LEXICMAP HSPS	WALKER20K OK	CLEAN_FULL	PERM. SIEVED
5.88 M	47.2%	6,827	79,621
24 GB	9,448 / 20k	pid 98.75	of 1.15 M



Shared spike reference on top, **RBD** brown. **Top:** kmindex → walker 20k. **Bottom:** LexicMap permissive. Columns align.
match A C G T gap

D *H. pylori*: 25-gene panel

ROUTE B · STREAMER

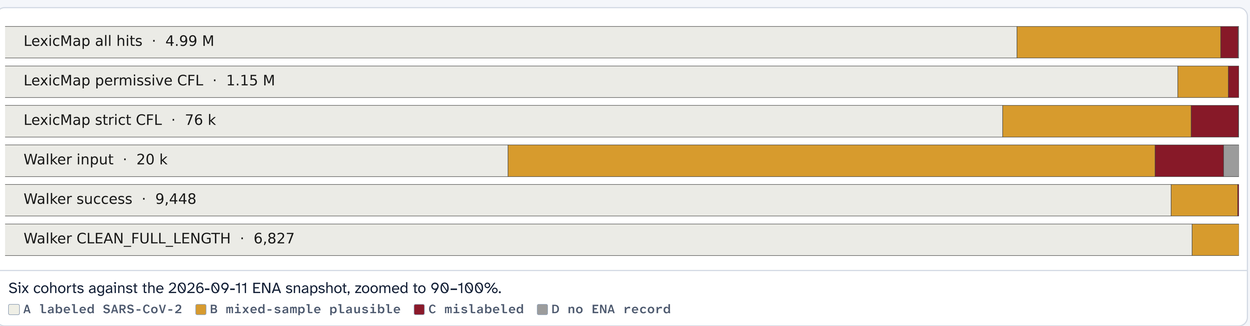
Four audit tiers per marker in one reference frame: clean full-length through low-coverage fragment, every accession accounted for.

STRICT MARKERS	LOOSE MARKERS	AUDIT TIERS	HYPHAEON PASS
20	5	4	oipA
conserved	selection	per marker	done



oipA strict panel: 1,042 clean accessions, sorted by substitution count.
match A C G T gap

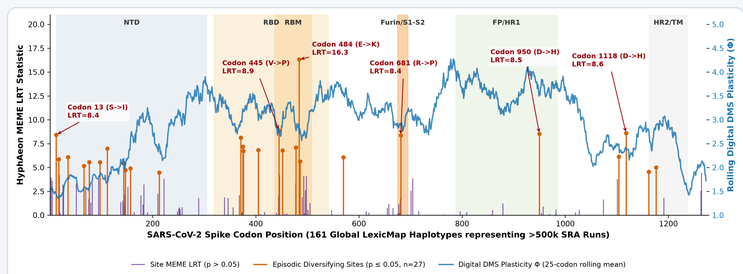
E SRA metadata: the walker drives contamination to zero



Six cohorts against the 2026-09-11 ENA snapshot, zoomed to 90–100%.
A labeled SARS-CoV-2 B mixed-sample plausible C mislabeled D no ENA record

WALKER20K INPUT	WALKER20K SUCCESS	CLEAN_FULL_LENGTH	LEXICMAP PERMISSIVE
0.555%	0.011%	0.000%	0.084%
C-mislabeled, raw hits	1 in ~9,500	0 / 6,827 mislabeled	no walker filter

F HyphAeon: selection on the recovered cohort



27 diversifying sites across 161 global spike haplotypes, led by antibody-escape hub **E484K**. Shaded bands mark spike domains.

HAPLOTYPES
161
>500k SRA runs

TOP SITE
E484K
MEME LRT 16.3

RUNTIME
29.5 s
no input tree

Logan unitigs → `kmindex` + `logan-walker` **Logan contigs** → `LexicMap` + `streamer` **sieved cohort** → **HyphAeon · ChronAeon · RhizAeon**

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Software & data: github.com/nekrut/disassembler · github.com/veg/HyphAeon · Zenodo archive: doi.org/10.5281/zenodo.23737199. Related CSHL posters: Galaxy UDTs (van den Beek) · Galaxy Workflow Foundry (Chilton) · HighGen (Burhans).

disassembler · Zenodo Data · HyphAeon